



Polymorphism

1. Generic pointer
2. Typecasting
3. Static and dynamic binding
4. Virtual functions
5. Implementing polymorphism
6. Real life examples

- Pointer of base class can point to objects of its derived class. Such base class pointer which is used for pointing to derived class objects is called as generic pointer.
- But pointer of derived class CAN NOT point to objects of its base class
- Even a pointer of one class can not point to objects of its peer level classes.
- Same is applicable for reference variables also. Such reference variables are called as generic references.

- Generic pointers can ONLY point to objects of derived class BUT they can't access members of derived class.
- To do so, we need to typecast them from base pointer to corresponding derived class pointer.
- Changing derived class pointer to base class pointer is called as Upcasting. This is always safe and successful.
- Changing base class pointer to derived class pointer is called as Downcasting. This can lead to runtime errors if the datatype of object doesn't support it. Hence it must be done by checking type of object at runtime. This is called as Runtime type identification (RTTI)

- Concept of function binding
- If the decision is done by compiler, based on data type of invoking entity, it is static binding (Also called as compile time binding)
- If the decision is done at run-time, based on data type of actual object, it is dynamic binding (Also called as run time binding)
- All static functions are statically bound
- All functions invoked using object, are statically bound
- All non-virtual functions are statically bound
- Only virtual function invoked using a pointer is dynamically bound

- A virtual function invoked using a pointer leads to dynamic binding
- Only non-static member functions can be virtual
- Static and global functions can not be declared as virtual
- A virtual function makes that class a Polymorphic class.
- In other words, a class having at least one virtual function present inside it directly or inherited from any of the base class, becomes Polymorphic class.
- Polymorphic class has Virtual table as its static data member and a virtual pointer as its non-static data member.
- So due to presence of virtual pointer, size of all objects of polymorphic class is 4 bytes more.

- Invoking a virtual function using a generic pointer leads to polymorphism.
- The message is same but due to dynamic binding, each function call is bound to different function based on data type of the object. This leads to different implementation of same message.
- Polymorphism can be achieved with generic reference also.

- Behavior of vehicles at traffic signal
- Operating multiple devices with same switch
- Saving data to different devices (like file or database etc)
- Encryption and decryption of data using different algorithms